

AI FUNDAMENTALS SERIES · PART 2 OF 3 · SECTIONS 06 TO 09

AI (LLM) Important Terms

Tokens, The Context Window, Embeddings, Semantic Search & Hallucinations

Instructor & Student Guide · 4 Core Operational Concepts · Production LLM Engineering

Part 2 Overview: This module explores the core operational concepts every developer and engineer encounters when interacting with Large Language Models: how text is chopped into tokens, how the context window limits working memory, how embeddings turn semantics into geometric coordinates for RAG, and why LLMs hallucinate with high confidence.

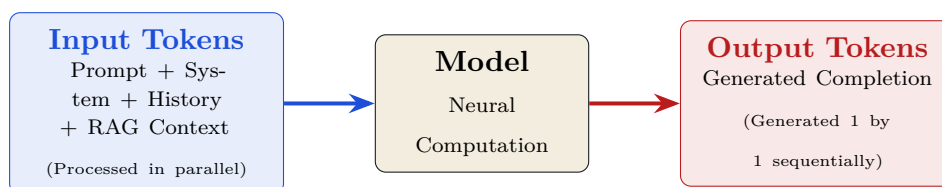
1 06. Tokens: How Models Read & The Economics of APIs

Models do not read words or characters. They process numbered subword chunks called **Tokens**, and all API billing, rate limits, and latency constraints operate strictly per token.

1.1 How Tokenization Works (Byte-Pair Encoding)

TOKENIZATION FUNDAMENTALS

ENGLISH RULE	1 Token $\approx$ 4 characters or $\approx$ 0.75 words (e.g., 1,000 tokens $\approx$ 750 words).
COMMON WORDS	Common words (e.g., “the”, “is”) form a single token.
RARE WORDS	Split into subwords (e.g., “unbelievable” $\rightarrow$ [“un”, “believ”, “able”]).
SPECIAL SCRIPTS	Code indentation, mathematical equations, and non-Latin scripts (Hindi, Japanese) consume substantially more tokens per sentence.



WHY OUTPUT TOKENS COST 3× TO 5× MORE THAN INPUT TOKENS

- **Input processing is parallel:** The GPU can ingest all 2,000 prompt tokens in a single matrix pass.
- **Output generation is sequential (Autoregressive):** To generate 200 tokens, the model must execute 200 complete forward passes through all 70B+ weights, waiting for token N before generating token $N + 1$.

2 07. The Context Window: Working Memory vs Permanent Storage

Definition: The Context Window is the maximum number of tokens a model can process in a single interaction. Everything outside this window does not exist for the model.

ALLOCATION INSIDE A 128K CONTEXT WINDOW:

1. System Prompt (Instructions)
2. Chat History
3. Retrieved Documents (RAG)
4. User Query
5. Output Budget

2.1 Context (Working Memory) vs Long-Term Memory (Storage)

Attribute	Context Window (Working RAM)	Long-Term Memory (External Database)
Persistence	Rebuilt fresh on every request; disappears on completion.	Persists indefinitely in PostgreSQL, Vector DB, or Redis.
Cost Model	You pay per token on every single call you re-send it.	Cheap disk storage; only queried when needed.
Failure Mode	When full, older tokens get truncated or summarized.	Scale is unlimited; requires query search infrastructure.

▪ THE WHITEBOARD VS LIBRARY ANALOGY

Think of the **Context Window** as a small whiteboard on the model's desk. It can only solve what fits on the whiteboard. When the whiteboard is full, you must erase earlier notes to write new ones. A **Vector Database** is the library down the hall — your code must fetch books from the library and write excerpts onto the whiteboard.

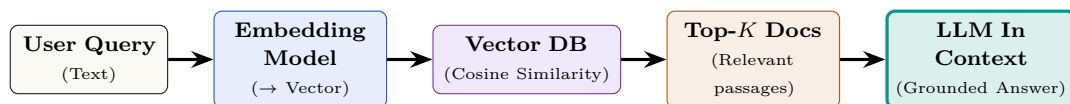
3 08. Embeddings: Turning Meaning into Coordinates

Definition: An embedding is a high-dimensional vector (array of numbers, e.g., 1,536 dimensions) representing the semantic meaning of a text passage.

SEMANTIC DISTANCE VS KEYWORD MATCHING

- "I love this movie" $\Rightarrow [0.21, -0.42, 0.87, \dots]$
- "This film was amazing" $\Rightarrow [0.19, -0.38, 0.91, \dots]$ $\rightarrow$ **Near identical distance!**
- "The weather is terrible" $\Rightarrow [-0.66, 0.12, -0.30, \dots]$ $\rightarrow$ **Far away in vector space!**

The first two share **zero common words**, yet sit right next to each other in embedding space.



WHY RAG (RETRIEVAL-AUGMENTED GENERATION) IS THE INDUSTRY STANDARD

RAG = Embeddings (Vector Search) + Context Window Injection

- No model retraining or fine-tuning required.
- Private enterprise documents never leak into public model weights.
- Answers cite specific source documents, making outputs verifiable and trustworthy.

4 09. When AI Sounds Confident but is Wrong: Hallucinations

***Fluency is not the same as Truth.** Language models are trained to optimize for **plausible continuation** conditioned on previous tokens. When facts are missing, the model generates whatever sounds statistically most natural.*

4.1 Next-Token Probability Sampling

When given: “The capital of Australia is ...”

- **Canberra:** 61% probability · **Sydney:** 23% · **Melbourne:** 9%

The model samples from this probability distribution. It does **not** consult a factual database inside its weights.

4.2 Raw LLM vs Grounded System

Dimension	Raw LLM (Ungrounded)	Grounded System (RAG / Tool-augmented)
Mechanism	Hallucinates plausible dates, names, and citations from weights.	Constrained to answer only from retrieved verified source documents.
Failure Mode	Inaccurate answers delivered with 100% confidence.	Explicitly replies: “Source documents do not contain this information.”

THE 5-LAYER RELIABILITY STACK FOR PRODUCTION AI:

1. **Grounding:** Supply external source material directly into context.
2. **Retrieval:** Fetch verified passages using semantic vector search.
3. **Citations:** Force the model to output clickable document references for every claim.
4. **Structured Validation:** Enforce strict JSON schemas (e.g., Pydantic/Zod) and unit tests.
5. **Human Review:** Mandatory human-in-the-loop for high-stakes medical, legal, or financial actions.

Module 2 Master Summary Spine

QUICK REVISION & MASTER TAKEAWAYS

